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Task Description:	The remarkable advances made in science, particularly in genomics and proteomics, over the last decade have enormous implications for the fields of evolutionary and gravitational biology. We are interested in how vascular plants, whose varied life forms support and nurture all terrestrial life, have evolved their complex, integrated, multidimensional networks for controlling and regulating the outcome of phenolic radical-radial coupling. These provide vascular plants with not only the ability to stand upright, and to sustain-or strive to attain-equilibrium positions for their stems and branches, but also the means for almost every essential developmental stage from seed to seed. It can thus be argued that evolution of the phenylpropanoid pathway was necessary for the successful and continued colonization of land by (vascular) plants. The sequencing of the Arabidopsis genome now offers an unprecedented opportunity in which to dissect and apprehend the diversity of phenolic radical-radical coupling and polymerization processes in planta. This includes not only identification of the cell type(s) (e.g. vessels, fibers, etc.) that utilize a particular coupling process, enzyme or protein at each and every stage of development, but also to gain insight into how these biochemical networks are modulated, e.g. in response to perceived environmental changes, such as the gravitational vector. A related phase of the study will identify the factor(s) engendering a shift in phenolic radical-radical coupling processes in loblolly pine during the re-alignment of its stems and branches for so-called compression wood formation. We are also interested in how the processes for assembly of secondary walls in vascular tissues of the seed plants evolved, particularly since they seem to differ from that of the primitive (extinct) early plants. Since various forms of "primitive" and so-called advanced plants are in existence today, questions surrounding cell wall assembly and differential evolution of the phenylpropanoid pathway can now be rationally examined using various molecular approaches and physico-chemical techniques. (i) to fully map out temporally and spatially, during the full life cycle of Arabidopsis thaliana, the networks that have evolved (both in a cell- and tissue-specific manner) for all dirigents, their homologues, putative lignin “scaffolding” proteins, and oxidative enzymes (e.g. laccases and peroxidases). This will provide the first comprehensive determination of the coordinated networking of phenolic coupling processes, and their cell specificities (i.e. vessels, fibers, etc.) for all stages of growth and development. (ii) to identify the factors responsible in gymnosperms for changes in designation of a cell from being a “normal” tracheary element to that of a “reaction” wood tracheary element. (iii) to define how primitive land plant cryptogams (Equisetum, Huperzia, Isoetes, Lycopodium, Psilotum, Selaginella) and the ferns (Ophioglossales and Filicales) undergo secondary wall thickening to generate the vascular apparatus, in terms of determining the mechanism(s) of biopolymer deposition, and the (bio)chemical basis of the phenolic coupling system(s) therein. (iv) to establish how the phenylpropanoid pathway (Phe monolignols) may have evolved during the transition to land, and in vascular plant evolution thereafter.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Multicellular vascular plant evolution from its unicellular aquatic forerunners was arguably one of the most important developments in evolutionary biology, since it ultimately provided the means for both the expansion and continued sustenance of all of life's resplendent terrestrial forms. The plant life that resulted markedly defines and characterizes our terrestrial ecosystems, and the conquest of land by plants also led to all of our modern soils and ultimately the varied life forms therein. It also provided the means to support the subsequent explosion of the terrestrial (metazoan) groups as they emerged. Moreover, vascular plant life markedly impacts our atmospheric chemistry and climate, including the levels of gaseous CO2. Humanity, of course, has long used (woody) plants for sundry purposes, e.g., as a source of fiber, solid wood products, foods, medicines, intermediate chemicals, flavors and fragrances, and for many other applications. Understanding the basis for multicellular vascular plant evolution is thus arguably crucial in defining how our ecosystem(s) and life forms came into existence. Understanding the basis for the differential formation of our various foods and fibers is also essential to the well-being and continued existence of the human race.
	Networks in Arabidopsis controlling/regulating phenolic coupling. We have established the complex pattern of expression of the 16-membered dirigent gene isovariant family in the vasculature apparatus (all tissues and organs), during each and every stage of Arabidopsis growth and development (Kim et al., manuscript in finalization). We also determined/identified the role of a specific (cationic) peroxidase isoenzyme in lignin assembly in the vascular tissue, as well as detecting specific cell-wall associated laccases, peroxidases, and H2O2-generating systems are present in lignifying vascular tissues. In short, the secrets of phenoxy radical coupling processes in cell wall assembly (as regards lignification, etc.) are being systematically teased apart. Dirigent proteins/proteins harboring dirigent sites: We are committed to defining exactly how vascular plants control and organize their varied phenoxy radical coupling processes in vivo. In that context, dirigent proteins (Latin: to guide or to align), discovered by our group are a new class of proteins which stipulate the outcome of radical-radical coupling processes. Significant attention was thus placed on delineating the presumed unique biochemical mechanism of the (+)-pinoresinol forming dirigent protein, with equilibrium binding assays and assays with varying levels of steady-state coniferyl alcohol radical concentrations [generated by action of laccases or peroxidases] establishing that they and not the parent phenols are the actual substrates. Thus the dirigent protein functions via a free-radical capture mechanism and this unique mode of biochemical action represents a new paradigm in biochemistry. In an analogous manner, evidence was obtained for control of other stereoselective phenylpropanoid (monolignol/hydroxycinnamic) acid coupling processes, as well as of regiospecific allylphenol coupling, e.g., in Larrea species. Work by others has also led to a much improved understanding of how regiospecific coupling occurs affording the structurally complex plant (polymeric) ellagitannins. Thus, in every biological system systematically examined thus far, strict biochemical control is exercised over phenoxy radical coupling, i.e., whether the products are optically active or not. [A similar situation now also holds for lignification.] Furthermore, based on immunolocalization data, we demonstrated the presence of dirigent proteins (for lignan biogenesis) in cambial cells/parenchyma cells of lignifying tissues, as well as evidence for dirigent protein-like epitopes in cell wall regions where lignin deposition is initiated; the latter data were further supported by in situ hybridization analyses. We thus proposed that lignification most likely involved participation of related proteins harboring arrays of

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dirigent sites (i.e., containing motifs for substrate radical binding) in order to permit ordered lignin biosynthesis. We therefore next examined to what extent there were functionally redundant dirigent (homologues) metabolic networks present in the vascular apparatus of Arabidopsis and which of these were coincident with lignification (Kim et al., manuscript in finalization).

Thus, each of the 16 dirigent isovariants of Arabidopsis, all with no introns, was isolated from genomic DNA together with their promoters (~700-2100 bp). The latter, with their 5¢ and 3¢ untranslated regions, were individually cloned into vectors together with gus or gfp reporter gene systems, then ligated into a p-CAMBIA 1305 vector, transformed into Agrobacterium tumefaciens (LBA 4404 strain GV2101) and finally Arabidopsis. The resulting T2 transformants (hygromycin-selected) for each line were individually assayed (e.g., for GUS activity) at each stage of growth and development until senescence.

As predicted, the results obtained revealed the presence of a comprehensive (cell and tissue-specific) network with each dirigent gene (homologue) having a unique pattern of expression in the vascular apparatus, thereby providing further insight into the presence of (partially overlapping) metabolic networks presumably controlling various aspects of phenoxy radical coupling. For example, 11 of the 16 dirigents, e.g., Dir5, 8, and 10 were expressed in the lignifying leaf vasculature, two at the base of lignifying leaf trichomes, four in the lignifying leaf hydathodes (vasculature), twelve in the lignifying abscission zone of the siliques, and twelve differentially expressed in lignifying regions of the stem vasculature [i.e., protoxylem, vascular cambium of xylary and interfascicular fiber, etc.]. Other overlapping, functionally redundant, metabolic networks were also observed in the monolignol forming pathway study, e.g., for cinnamyl alcohol dehydrogenase, (Kim et al., manuscript in finalization), 4-coumarate CoA ligase, 22 phenylalanine ammonia lyase and cinnamoyl CoA reductase23 (Cochrane et al., manuscripts in preparation) multigene families.

Oxidative processes in lignifying cell walls. In addition to the dirigent multigene network, it was important to delineate how monolignol substrates were oxidized and how H2O2-generation was occurring in lignifying cell walls. In this context, a lignification-associated cationic peroxidase TP60 was isolated from tobacco, whose suppression led to transformants with lignin reductions up to 40-50% of wild type (control) plants, i.e., the highest level of lignin downregulation obtained thus far in any plant species. This, therefore, significantly helped in identifying part of the biochemical system responsible for free-radical substrate (monolignol) oxidation prior to lignin assembly. Additionally, mass spectrometric proteomic analysis of the Arabidopsis xylary elements/interfascicular fiber and epidermis/cortex tissues from 5 week old basal stem sections revealed that there were 11 peroxidases present, ten in all tissues examined and one unique to lignifying tissues While most of these peroxidases have high homology (>50%) to that of tobacco TP60, the role of each of the 11 putative cell wall peroxidases nevertheless still remains to be elucidated and thus is the subject of future work.

In an analogous manner, there are 17 putative laccases in Arabidopsis, and proteomic analysis of the five week old basal stem tissue revealed the presence of 5 in the lignifying vascular tissues, with none in non-lignifying tissues. Although there is substantial debate as to the possible role of laccases (directly or indirectly) in lignification, we need to establish their actual roles. As for the dirigent gene (homologue) family, the promoters of each laccase were obtained, fused to gus and gfp reporter systems, and then used to transform Arabidopsis as before (work in progress), in order to establish the comprehensive patterns of expression at all stages of growth/development. Additionally, the 5 laccases detected in lignifying tissues are currently being expressed in Aspergillus nidulans, in order to assess their biological (substrate specificity) properties.

H2O2-generating systems: Peroxidases require H2O2 as a cofactor, and in this regard we have isolated three copper amine oxidases present in lignifying vascular tissues (ongoing work), that are putatively involved in H2O2 generation in the various cell wall types present.

In summary, much progress has been made in systematically mapping out the processes operative in the vasculature during cell wall development, as regards the biochemistry of phenoxy radical coupling processes and related systems.

To define further how primitive cryptogams evolved their vascular apparatus, the (bio)chemical basis of their phenolic coupling system(s) and how the phenylpropanoid pathway evolved during plant adaptation to land. Primitive land plant cryptogams have survived on earth for more than 400 million years, and are generally regarded as evolutionary forerunners of the gymnosperms and flowering angiosperms. This phase of the work was thus directed towards gaining broad insight into how the vascular apparatus and phenylpropanoid pathway evolved during plant adaptation to land, and the findings made are in keeping with the preceding observations. First, ultrastructural studies were carried out on various members of the Lycopsida, (Huperzia lucidula, Lycopodium tristachyum, Selaginella kraussiana, and Isoetes lacustris), the Sphenopsida (i.e., Equisetum telmeteia), the Psilopsida (e.g., Psilotum nudum), as well as the Pteropsida (i.e., Ophioglossum reticulatum, Botrychium virginianum, Pteridium aquilinum and Blechnum spicant), i.e., representatives of the cryptogams. Briefly, anatomical studies, using phloroglucinol-HCl, for staining (apparent) guaiacyl lignified elements, suggested that phenolics were deposited in the vascular apparatus of all species examined, with the most intense staining observed in the later-forming metaxylem elements, relative to lighter staining in protoxylem (data not shown).

Second, the lignin contents/compositions were examined for each of the four basal tracheophyte classes (data not shown). Acetyl bromide analyses revealed a trend of apparent increasing lignin contents (of extractive-free cell wall residues), with lowest amounts (~3.5%) in the Lycopsida (most primitive class) and (~6%) in the Sphenopsida versus higher levels (~20-30%) in the Psilopsida and Pteropsida (highest taxonomic class). Thioacidolytic and nitrobenzene oxidation analyses next established that guaiacyl lignins were present in the Psilopsida and Pteridophyta, with much lower levels in the Lycopsida [e.g., H. lucidula]. Thus, in the basal tracheophytes, H-lignin formation apparently first evolved, this presumably representing the earliest type of lignin. As further elaboration of cell wall thickening/types occurred, guaiacyl (G) units were next formed and were differentially deposited, relative to the initial phase, as discussed earlier. By contrast, and as expected, none of the primitive plants generated S-lignins which apparently mainly appeared in the flowering angiosperms.

We also conducted a comprehensive analysis of the soluble phenolic metabolites present in the aforementioned species, with ~55 compounds being identified (Amakura et al., manuscript in preparation). In the Lycopsida, most were ferulic acid-derived, which contains the same aromatic (G) nucleus as coniferyl alcohol, this being consistent with the pathway extending beyond p-coumaryl alcohol to coniferyl alcohol. In the Sphenopsida and Psilophyta species examined, the metabolites were mainly caffeic and p-coumaric acid derivatives, whereas they were also caffeic acid and coniferyl alcohol derived in the Pteridophyta. Together, these data are consistent with stepwise progression of the phenylpro-panoid pathway from first forming H lignin/related metabolites, then to further hydroxylation/ methylation to

	caffeic acid derivatives and G lignin/related metabolites, respectively. Thirdly, in terms of evolutionary progression, we conducted an extensive survey as to whether phenylpropanoid pathway genes to the monolignols, ranging from the entry point of phenylalanine ammonia lyase (PAL) to cinnamyl alcohol dehydrogenase/dirigents, were in these basal tracheophytes, as well as in representatives of algae (Chara corallina), mosses (Physcomitrella patens), liverworts (Pellia epiphylla), hornworts (Phaeoceros laevis), and cycads (Cycas revolute). As expected, all species examined harbored multigene PAL families,28 these being =50% identical to Arabidopsis PAL at the amino acid level; analysis of these data also gave interesting insights vis-à-vis PAL orthologs and paralogs. Additionally, p-coumarate 3-hydroxylase (C3H), which introduces the second hydroxyl group on the aromatic ring to ultimately give entry into the guaiacyl metabolites, was present in all organisms examined and with high homology (76-94% similarity) to the Pinus taeda homologue. Work is currently underway to complete analysis of the evolutionary trends in this pathway, including the later stages, i.e., with cinnamoyl CoA reductase, cinnamyl alcohol dehydrogenase, and the dirigent homologue families. [In those cases, the corresponding recombinant proteins are being examined in order to establish substrate specificities/ versatilities for each, and thus to explain biochemically how the pathway differentially evolved to form both H and G moieties.] Taken together, this phase of the study has proceeded smoothly, and has provided a comprehensive understanding of how the phenylpropanoid pathway evolved in algae, primitive cryptogams, gymnosperms, and flowering angiosperms, this affording a much improved understanding of how land plant adaptation and elaboration of the lignin forming processes differentially occurred over time.
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